ADESH PANDEY

+1-934-386-0986 | adesh0805@gmail.com

Professional Summary

Geophysicist with advanced training and industry experience in seismic imaging, inversion, machine learning, distributed acoustic sensing (DAS), and hydrocarbon exploration. Experience spans computational geophysics, seismic interpretation and reservoir characterization, seismic survey design, quantitative analysis, and high-performance computing on CPU and GPU systems. These responsibilities require advanced knowledge of geophysics, applied mathematics, and computational modeling typically obtained through graduate-level education. Particularly qualified through doctoral research in geophysics, prior industry experience in exploration geophysics, and current technical work in advanced seismic modeling, inversion, and computational algorithms for subsurface characterization.

Post-Secondary Education

Colorado School of Mines **01/03/2022 – 12/19/2025**
Golden, Colorado, USA
Ph.D. in Geophysics
Graduate Certificate in Data Science and Machine Learning
GPA: 4.0/4.0
Harold Mendenhall Geophysics Award for best thesis

Indian Institute of Technology, Kharagpur **07/01/2010 – 05/31/2015**
Kharagpur, India
B.S. and M.S. in Exploration Geophysics
CGPA: 8.68/10.00
Institute Silver Medallist

Professional Experience

Geophysicist Graduate **01/20/2026 – Present**
Upstream R&D, BP America Inc.
501 Westlake Park Boulevard, Houston, TX 77079

- Apply advanced numerical methods and computational algorithms to perform seismic wave propagation modeling and inversion for subsurface characterization.
- Execute full waveform inversion (FWI) workflows to iteratively update subsurface models by minimizing misfit between observed and simulated seismic data.
- Conduct finite difference modeling for acoustic and elastic wave propagation in complex geological media.
- Perform seismic imaging workflows to generate high-resolution subsurface images for interpretation and validation.
- Conduct diving-wave illumination modeling to evaluate shallow subsurface sensitivity in complex geologic settings.
- Support seismic survey design through evaluation of acquisition geometry, subsurface illumination coverage, and data quality optimization.
- Perform Distributed Acoustic Sensing (DAS) illumination modeling for fiber-optic based seismic acquisition systems.
- Execute computationally intensive modeling and inversion workflows on high-performance computing environments using CPU and GPU systems.
- Apply numerical optimization techniques and scalable algorithms for large-scale geophysical modeling workflows.

Tools, software, technologies, and skills used:

- High-performance computing (HPC) on CPU and GPU systems
- Seismic inversion, FWI, finite difference modeling, seismic imaging
- DAS illumination modeling and seismic survey design
- Python, CUDA, C++

- Linux-based computational environments

Geophysical Technology Intern **05/01/2024 – 08/31/2024**
BP America Inc.
501 Westlake Park Boulevard, Houston, TX 77079
Project: *Automated Seismic Events Detection and Classification in Downhole Distributed Acoustic Sensing (DAS) Data Using Deep Learning*

- Developed deep learning-based computer vision workflows for automated detection and classification of seismic events in DAS data.
- Automated large-scale data analysis, reducing event detection timelines from months to hours.
- Integrated DAS data with petroleum engineering datasets to enhance wellbore condition analysis.
- Identified sanding and microseismic activity to support safe and efficient operations.
- Evaluated feasibility of real-time deployment of machine learning models for continuous monitoring.

Tools, software, technologies, and skills used:

- Python, TensorFlow, Keras, PyTorch
- DAS data analysis, machine learning, computer vision
- Signal processing and time-series analysis

Earth Science Intern **05/01/2023 – 08/31/2023**
Chevron Technology Center
1500 Louisiana St (HOU150), Houston, TX 77002
Project: *Machine Learning Assisted Seismic Inversion*

- Developed supervised machine learning workflows to predict rock properties (AI, Vp/Vs) from AVA attributes.
- Improved reservoir characterization accuracy and reduced cycle time through ML-assisted inversion.
- Performed comparative analysis between ML-based inversion and conventional simultaneous seismic inversion methods.
- Applied trained models to offshore basin datasets to support exploration decisions.
- Supported de-risking of near-field exploration opportunities using data-driven subsurface analysis.

Tools, software, technologies, and skills used:

- Python, Scikit-Learn, TensorFlow, PyTorch
- Seismic inversion, AVA analysis, reservoir characterization
- Machine learning model development, validation, and comparison

Lead Exploration Geophysicist **07/01/2015 – 12/31/2022**
Cairn Oil and Gas, Vedanta Ltd.
ASF Center, Tower A, Jwala Mill Road, Phase IV, Udyog Vihar Sector 18, Gurugram, Haryana 122016, India

- Delivered regional to prospect-level structural and stratigraphic interpretation in the Barmer Basin using more than 3,000 sq. km of 3D seismic data, 3,500 line-km of 2D seismic data, and over 180 exploration and appraisal wells.
- Modeled and mapped fault-dominated structural and stratigraphic plays, turbiditic channel-fan complexes, and basaltic and sub-basaltic systems, contributing to multiple exploration prospects.
- Integrated regional knowledge, field data, seismic attributes, quantitative interpretation, petrophysics, and rock physics models to mature and de-risk prospects and evaluate hydrocarbon resources.
- Supported drilling operations for exploration targets through seismic interpretation, well ties, seismic attribute studies, rock physics, and AVO analysis.
- Stewarded survey design, acquisition, processing, and quality control for more than 3,000 line-km of 2D seismic and 2,000 sq. km of 3D seismic data for hydrocarbon prospectivity assessment in new OALP blocks.
- Led acquisition and processing of more than 15,000 line-km of Full Tensor Gravity (FTG) data in the Barmer Basin and used the results to optimize seismic survey design.
- Led technical data rooms with interested companies by presenting raw and interpreted data and prospect overviews for geologic and geophysical evaluations.

- Managed a portfolio of exploration leads and prospects for potential new hydrocarbon pools.

Notable projects and accomplishments:

Geophysics Intern **05/01/2014 – 06/30/2014**
Schlumberger Asia Pvt. Ltd.
Mumbai, India

Project: *Processing and Post-Drill Statistical Analysis of Zero Offset Vertical Seismic Profiling (ZVSP) Data*

- Processed and performed inversion of ZVSP data to quantify uncertainty between predicted and actual P-wave data.
- Supported look-ahead analysis and informed drilling decisions through post-drill statistical evaluation.

Geophysics Intern **05/01/2013 – 06/30/2013**
Oil and Natural Gas Corporation (ONGC)
Kolkata, West Bengal, India

Project: *3D Seismic Data Acquisition and Processing*

- Analyzed up-hole survey data to determine optimum depth and prepared near-surface models for shot processing.
- Assisted in designing acquisition parameters for 3D seismic data acquisition.

Awards and Honors

- Harold Mendenhall Geophysics Award, Colorado School of Mines, 2026
- SEG Foundation Scholarship, 2023–2025
- Chevron Fellowship, 2023–2024
- Second Place, Oral Presentation, Graduate Research and Discovery Symposium, Colorado School of Mines, 2023
- Best Poster Award and People's Choice Award, Technical Forum, Cairn Oil and Gas, 2018
- Institute Silver Medal, IIT Kharagpur, 2015
- K.L. Chopra Award, IIT Kharagpur, 2015
- P.K. Bhattacharya Award, IIT Kharagpur, 2015

Peer-Reviewed Journal Publications

Research Metrics

- Citations: 20+ spanning top journals (Scientific Reports, Seismica, Optical Fiber Technology, Geophysical Journal International, GEOPHYSICS)

Peer Reviewed Articles

- Davis, D., Shragge, J., de Ridder, S., Girard, A. J., & Pandey, A. (2026). Interferometric frequency-domain gradiometry. *Geophysical Journal International*, 245(3), ggag146.
- Pandey, A., Shragge, J., & Girard, A. J. (2025). First-order control factors for ocean-bottom ambient seismology interferometric observations. *Geophysical Journal International*, 243(2), ggf351.
- Pandey, A., de Ridder, S., Shragge, J., & Girard, A. J. (2025). Ocean-bottom seismic interferometry in coupled acoustic-elastic media. *Geophysical Journal International*, 242(3), ggaf271.
- Pandey, A., Shragge, J., Chambers, D., & Girard, A. J. (2023). Developing a distributed acoustic sensing seismic land streamer: Concept and validation. *GEOPHYSICS*, WC59–WC67.

Peer-Reviewed Manuscripts Under Review

- Pandey, A., et al. (2026). Scholte-wave adjoint tomography for building low-frequency offshore shear-wave velocity models. Under review in *GEOPHYSICS*.
- Kumar, A., Sethi, H., Tsvankin, I., & Pandey, A. (2026). Regularizing elastic full-waveform inversion using a conditional denoising probabilistic model. Under review in *GEOPHYSICS*.

Conference Publications and Oral Presentations

- Pandey, A., Shragge, J., & Girard, A. J. (2024). Ambient cross-correlation modeling for dense ocean-bottom node arrays: A Gulf of Mexico example. *SEG/IMAGE Annual Meeting*, Houston, Texas.

- Pandey, A., Dasgupta, S., Kuila, U., Yadav, R. K., Mishra, P., & Mohapatra, P. (2017). Various Characteristics of Deccan Lava Flow Sequences Identified in Core, Wireline and Seismic Data in the Barmer Basin. *Society of Petroleum Geophysicists (SPG) Conference*, Jaipur, India.
- Pandey, A., Dasgupta, S., Kuila, U., Yadav, R. K., & Mishra, P. (2019). Multi-Scale Volcanic Facies Characterization of Deccan Volcanic Complex in the Barmer Basin of Rajasthan: Implications for Exploration in a Flood Basalt Province. *AAPG International Conference*, Cape Town, South Africa.

Conference Poster Presentations

- Pandey, A., et al. (2026). Scholte-wave adjoint tomography for building low-frequency offshore shear-wave velocity models. *SEG/IMAGE Annual Meeting*, Houston, Texas. .
- Pandey, A., et al. (2025). First-order Control Factors for Ocean-bottom Ambient Seismology Interferometric Observations. *SEG/IMAGE Annual Meeting*, Houston, Texas..

Professional Service

Journal Peer Reviewer

- Completed **40+ invited peer reviews** for leading international journals in geophysics, remote sensing, and subsurface imaging, including:
  - *GEOPHYSICS* – 5 reviews
  - *Journal of Applied Geophysics* – 6 reviews
  - *IEEE Transactions on Geoscience and Remote Sensing* – 2 reviews
  - *Interpretation* – 4 reviews
  - *Journal of Geophysical Research* – 1 review
  - *Tunnelling and Underground Space Technology* – 2 reviews
  - SEG conference abstracts – 20+

Conference Judging and presentations

- Served as an invited **Technical Judge** and **Session Chair** for **three** Society of Exploration Geophysicists (SEG/IMAGE) conferences, evaluating technical presentations based on scientific merit, technical innovation, and research impact, moderating presentations and facilitating scientific discussions among researchers and industry experts.